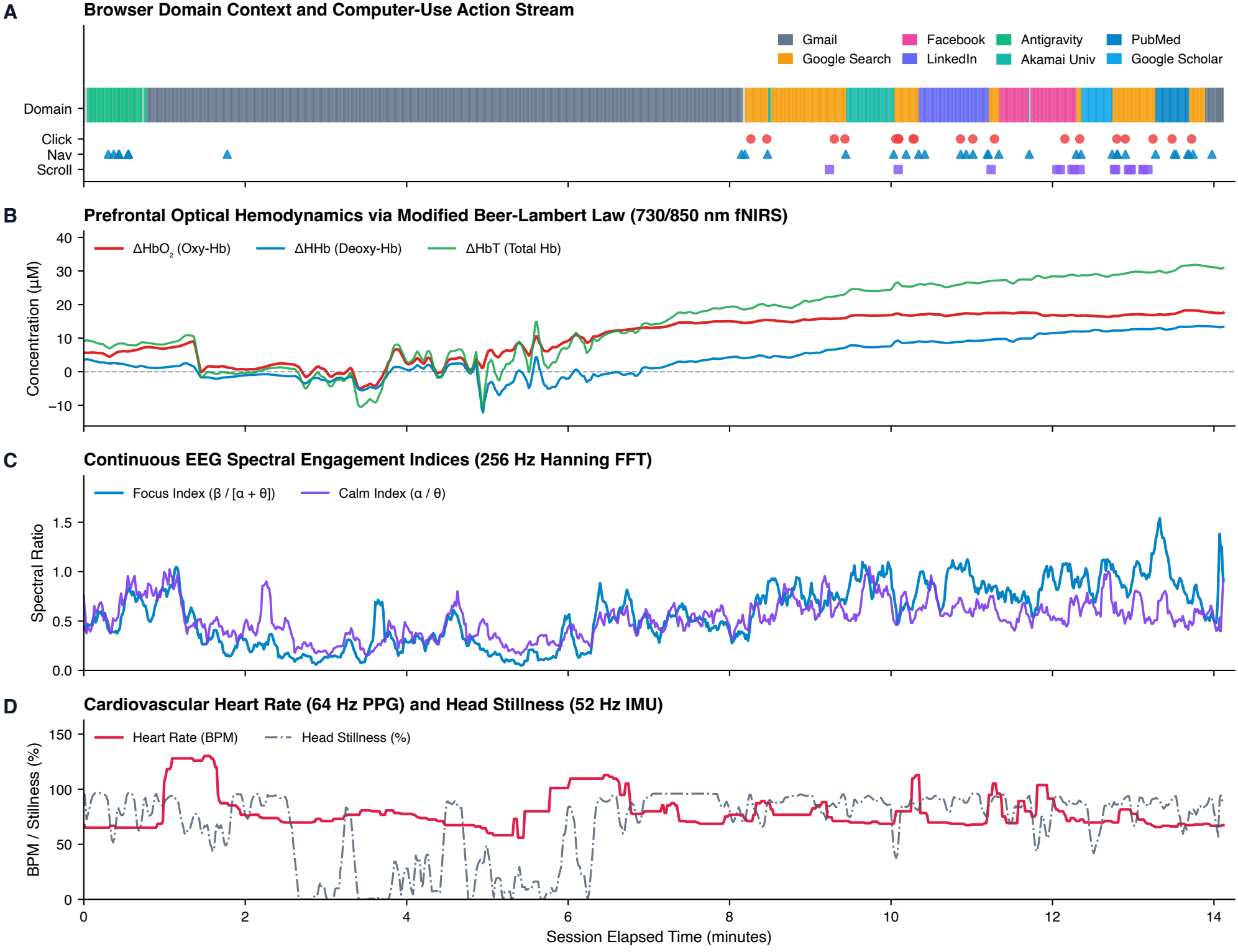

A
Browser Domain Context and Computer-Use Action Stream
Gmail
Google Search
Facebook
LinkedIn
Antigravity
Akamai Univ
PubMed
Google Scholar
Domain
Click
Nav
Scroll
B
Prefrontal Optical Hemodynamics via Modified Beer-Lambert Law (730/850 nm fNIRS)
ΔHbO2 (Oxy-Hb)
ΔHHb (Deoxy-Hb)
ΔHbT (Total Hb)
Concentration (μM)
40
30
20
10
0
−10
C
Continuous EEG Spectral Engagement Indices (256 Hz Hanning FFT)
Focus Index (β / [α + θ])
Calm Index (α / θ)
Spectral Ratio
1.5
1.0
0.5
0.0
D
Cardiovascular Heart Rate (64 Hz PPG) and Head Stillness (52 Hz IMU)
Heart Rate (BPM)
Head Stillness (%)
BPM / Stillness (%)
150
100
50
0
0
2
4
6
8
10
12
14
Session Elapsed Time (minutes)